



GTPI302 Guided Learning My Learning Contract

to be included in E-Portfolio
(revised on Oct 2025)

Name of Student:	Abdul Hamid	Admin No.:	2503696C									
Name of Learning Facilitator:	Roselind Nah											
Project Title:	Web Development with Python and JavaScript											
Brief Description of your project	What am I going to do? This project focuses on learning web app development through a structured, hands-on approach. By following CS50W lectures on edX, I will study core technologies such as HTML, Python, JavaScript, and SQL. Weekly lectures will be paired with practical problem sets to reinforce concepts, while additional support will come from GenAI tools and online communities. The goal is to build both theoretical understanding and practical skills in modern web development and database management.											
My Learning Goals	Refine Learning Goals as needed: <i>E.g. to work on my marketing skills.</i> - Master Core Web Technologies - Develop Full-Stack Problem-Solving Skills - Build Independent Learning and Collaboration Habits My project will help me cultivate these TP Student Profile attributes (choose three attributes) * Refer Appendix I for each TSP description <table border="0"> <tr> <td>☐ Inquisitive</td> <td>☐ Integrity</td> <td>☒ Critical thinking</td> </tr> <tr> <td>☐ Resilient</td> <td>☐ Empathy</td> <td>☒ Digitally Fluent</td> </tr> <tr> <td>☒ Growth-Oriented</td> <td>☐ Communicates Effectively</td> <td>☐ Innovative & Entrepreneurial</td> </tr> </table> For each chosen attribute, how would you rate yourself on a scale of 1-5 at this point (e.g. in the context of school work, personal life, part-time jobs)? 1: Never used or exhibited this attribute 2: Rarely using or exhibiting this attribute 3: Sometime using or exhibiting this attribute 4: Most of the time using or exhibiting this attribute 5: Always using or exhibiting this attribute For each chosen attribute, provide your current rating and explain how you can develop each attribute through GL. Critical Thinking: I will strengthen my ability to analyze problems, debug code, and design effective solutions in web development. Digitally Fluent: I will build fluency in essential digital tools and programming languages such as HTML, CSS, Python, JavaScript, and SQL. I will also learn how to host a website on server with backend programming Growth-Oriented: I will cultivate continuous learning habits by engaging with structured lectures, problem sets.			☐ Inquisitive	☐ Integrity	☒ Critical thinking	☐ Resilient	☐ Empathy	☒ Digitally Fluent	☒ Growth-Oriented	☐ Communicates Effectively	☐ Innovative & Entrepreneurial
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Target beneficiary/ beneficiaries (if applicable):	N/A		
Plan your learning; how do you achieve your own learning goals?	What do I already know/do in this area? I already have a foundation in Python basics and have explored CS50x. I have practical experience developing web applications using Flask (a Python framework) and SQLite3, leveraging the CS50 library as a beginner-friendly tool to manage databases and workflows.		
	What new knowledge & skills will I need to complete this project? New knowledge needed: HTML, CSS, JavaScript, web architecture, HTTP, APIs, advanced SQL, security principles and backend programming. New Skills needed: Full-stack application building, Git/GitHub version control, web app deployment, debugging and problem-solving		
	What possible challenges would I need to overcome for my learning project? Time Management: Each lecture is ~2.5 hours, with problem sets requiring several hours. Overcome by: Creating a precise, feasible weekly schedule to balance workload. Lack Traditional Classroom Support: No direct team to guide me. Overcome by: Seeking help from GenAI tools, friends from IIT, Lecturers from TP and online communities. Outdated Content: Some course material may not be up to date . Overcome by: Researching and applying up-to-date practices when I encounter outdated approaches.		
	I plan to get help from: Please tick and name the specific organisation(s)/expert(s) you would consult, include designation, names, telephone numbers and email ☐ Partner departments in TP: _____ ☐ Partner Schools in TP: _____ ☐ External Partner Organisation: _____ ☐ External Resource People: _____ ☒ Others: <u>CS50's community</u>		
	I plan to use these learning resources: Please tick and name the resources you would use. ☐ Books/newspapers: _____ ☒ Online learning/courses: <u>CS50's Web Programming with Python and JavaScript</u> ☐ Internet journals/articles: _____ ☐ Experiential learning: _____ ☐ Documentaries/videos: _____ ☐ Others (e.g. seminars, interviews, performances, lectures, workshops) _____		
Action Plan Timelines &	Week	Actions	Deliverables
	Plan for the entire semester	Specify the actions you will be taking to complete your project	Specify concrete and tangible deliverables you will produce

Deliverables	on a weekly basis		throughout the semester
	Week 1	Contact my facilitator and discuss the action plan	
	Week 2	Watch Week0 and Week1 Lecture	Work On Project 0
	Week 3	Watch Week 2 Lecture	Work On Project 0
	Week 4 (Flex Week 1)	Complete Project 0	
	Week 5	Watch Week 3 Lectures	Work on Project 1
	Week 6	Watch Week 4 Lectures	Work on Project2
	Week 7	Complete Project 1 and 2	
	Week 8	Complete Project 1 and 2	
	Week 9 (Term Test Week)		
	Week 10 (Term Break)		
	Week 11 (Term Break)		
	Week 12	Watch Week 5 Lectures	Work on Project3
	Week 13	Watch Week 6 Lectures	Work on Project3
	Week 14	Watch Week 7 Lectures	Work on project3
	Week 15 (Flex Week 2)	Watch Week 8 Lecture	
	Week 16	Complete E portfolio	Work on project 4
	Week 17	Submit E portfolio	
	Please customise if your project commences prior to the regular semester timeframe.		
	Plan your E-portfolio; Learning evidence for E-portfolio	What do you include in your E-portfolio? e..g. Report, prototype, checklist, glossary of terms, reflection, draft designs, surveys, survey analyses, interview transcripts, etc. (50-100 words).	
My E-portfolio will include a weekly report summarizing what I learned from each lecture and problem set, along with reflections on challenges and solutions. I will showcase my practical work by uploading coded projects, either as functioning websites or code files, to demonstrate applied skills			

Annex 1

TP Student Profile attributes

Lifelong Learner

TSP	TSP Attributes
Inquisitive	Being curious, explore, question, and seek knowledge across disciplines.
Resilient	Adapt, persevere through challenges, and thrive in changing environments.
Growth-oriented	Seek continuous improvement and take ownership in learning.

Values-centered Leader

TSP	TSP Attributes
Integrity	Act with ethics, honesty and strong moral principles.

Empathy	Understand and respect diverse perspectives with compassion and global awareness.
Communicates Effectively	Express ideas clearly and engage meaningfully with others.

Future-oriented Creator

TSP	TSP Attributes
Critical Thinking	Analyse information objectively to make informed decisions.
Digitally Fluent	Navigate digital tools, technologies, and information discerningly.
Innovative & Entrepreneurial	Create new ideas, take initiatives, take calculated risks, and embrace opportunities.